

**M05 / PROTECTED ANSWERS**

# M05 Explained Reviews

**Standalone Zero-to-Job-Ready Data Engineer 2026**

Study, practise, check and explain.

**A complete learning route**

Start with the lesson video. Read the worked explanation beside it, pause to reproduce the example, then solve the supplied task. The books contain your study notes; keep your own work and results as evidence.

Use the protected answers after saving your first attempt. A changed retry shows what you can now do independently. There is no forced timer during learning.

One course. M00-M19. 264 lessons and 50 project phases.

# Contents

Tap a topic or use the PDF bookmarks. These numbers are the actual pages in this edition.

|                                                                                    |   |
|------------------------------------------------------------------------------------|---|
| Read this first . . . . .                                                          | 3 |
| YOUR STANDALONE REVIEW BOOK . . . . .                                              | 3 |
| Open only after a saved attempt. . . . .                                           | 3 |
| Contents . . . . .                                                                 | 3 |
| Review DE05-01   CSV properly: parse, inspect, preserve raw . . . . .              | 3 |
| Review DE05-02   JSON and JSONL from zero . . . . .                                | 4 |
| Review DE05-03   Paths, folders and deterministic batch files . . . . .            | 4 |
| Review DE05-04   pandas profiling before transformation . . . . .                  | 4 |
| Review DE05-05   Cleaning with pandas and rejected rows . . . . .                  | 4 |
| Review DE05-06   pandas merge, cardinality and aggregation . . . . .               | 5 |
| Review DE05-07   Data validation: record, batch and relationship rules . . . . .   | 5 |
| Review DE05-08   HTTP and REST APIs from zero . . . . .                            | 5 |
| Review DE05-09   Calling APIs safely with requests . . . . .                       | 5 |
| Review DE05-10   Pagination and authentication without leaking secrets . . . . .   | 6 |
| Review DE05-11   Retries, rate limits and partial failure . . . . .                | 6 |
| Review DE05-12   Python + PostgreSQL with SQLAlchemy . . . . .                     | 6 |
| Review DE05-13   Configuration, environment variables and secret hygiene . . . . . | 6 |
| Review DE05-14   Logging, testing and reusable pipeline code . . . . .             | 7 |
| Review DE05-15   Integrated Python data-engineering pipeline . . . . .             | 7 |

## Read this first

Open this book after saving your own attempt. Compare the reasoning and result, repair the cause of a mistake, then close the answer and try changed inputs.

## YOUR STANDALONE REVIEW BOOK

### Open only after a saved attempt.

Use this book to diagnose a saved first attempt. Compare reasoning, observed state and controls rather than wording. Record the weakness, close this review, then solve the changed retry independently. Work done while review is open is repair evidence, not mastery evidence.

## Contents

|                                                                          |    |
|--------------------------------------------------------------------------|----|
| Review DE05-01   CSV properly: parse, inspect, preserve raw              | 3  |
| Review DE05-02   JSON and JSONL from zero                                | 3  |
| Review DE05-03   Paths, folders and deterministic batch files            | 4  |
| Review DE05-04   pandas profiling before transformation                  | 4  |
| Review DE05-05   Cleaning with pandas and rejected rows                  | 5  |
| Review DE05-06   pandas merge, cardinality and aggregation               | 6  |
| Review DE05-07   Data validation: record, batch and relationship rules   | 6  |
| Review DE05-08   HTTP and REST APIs from zero                            | 7  |
| Review DE05-09   Calling APIs safely with requests                       | 7  |
| Review DE05-10   Pagination and authentication without leaking secrets   | 8  |
| Review DE05-11   Retries, rate limits and partial failure                | 8  |
| Review DE05-12   Python + PostgreSQL with SQLAlchemy                     | 9  |
| Review DE05-13   Configuration, environment variables and secret hygiene | 9  |
| Review DE05-14   Logging, testing and reusable pipeline code             | 10 |
| Review DE05-15   Integrated Python data-engineering pipeline             | 10 |

### Review DE05-01 | CSV properly: parse, inspect, preserve raw

OPEN ONLY AFTER a saved genuine attempt.

Expected control: the module CSV has 9 data records and seven header fields. Notebook, A5 must remain one product value. A correct checksum is evidence of exact bytes, not correctness of business content. If your parser produced eight fields on that row, you used text splitting instead of CSV parsing.

Changed retry: Create a new CSV with a quoted city value Delhi, NCR and an ID 0009. Parse it without losing field boundaries or leading-zero identity.

Move on only when the changed retry works with this review closed and you can explain the failure/validation path.

## Review DE05-02 | JSON and JSONL from zero

OPEN ONLY AFTER a saved genuine attempt.

The nested JSON is one batch document containing 3 orders. The JSONL fixture is 3 independent event records. `lines=[]` is a known empty list; `customer=null` means the customer object is absent. Do not conflate document grain with order/event grain.

Changed retry: Create a three-line JSONL file where the second event has an optional nested metadata object. Process each line and explain compatibility without forcing all events to identical shape.

Move on only when the changed retry works with this review closed and you can explain the failure/validation path.

## Review DE05-03 | Paths, folders and deterministic batch files

OPEN ONLY AFTER a saved genuine attempt.

A defensible route anchors paths deliberately, sorts or selects inputs by an explicit rule, fingerprints inputs and refuses silent overwrites. If changing the terminal working directory breaks the script, path resolution is not yet reproducible.

Changed retry: Use filenames whose lexical order conflicts with a `business_date` inside each file. Implement and explain the intended ordering rule.

Move on only when the changed retry works with this review closed and you can explain the failure/validation path.

## Review DE05-04 | pandas profiling before transformation

OPEN ONLY AFTER a saved genuine attempt.

Profiling should describe the raw population before correction. The fresh inventory has 6 rows; one zero quantity and one nonnumeric `unit_cost` are quality risks. Reporting a cleaned 4-row profile as if it were raw hides the source condition.

Changed retry: Profile a changed file containing a blank numeric field and one duplicate key; do not clean it. Produce only observations and suspected risks.

Move on only when the changed retry works with this review closed and you can explain the failure/validation path.

## Review DE05-05 | Cleaning with pandas and rejected rows

OPEN ONLY AFTER a saved genuine attempt.

Authoritative controls for the supplied module fixture: 9 raw records; 4 accepted; 5 rejected; accepted amount 5100.00; accepted IDs MO01, MO05, MO06 and MO07. Both MO02 occurrences

are rejected because the batch contains two records with the same order\_id and provides no timestamp/version rule that would justify choosing one as authoritative. This is a deliberate conservative policy, not a universal deduplication rule. Reconcile raw = accepted + rejected and recompute 5100.00 independently from accepted quantity x unit\_price.

Changed retry: Use the fresh inventory fixture: valid unique SKU, positive integer quantity, nonnegative decimal cost and ISO received date. Split accepted/rejected with reasons.

Move on only when the changed retry works with this review closed and you can explain the failure/validation path.

## Review DE05-06 | pandas merge, cardinality and aggregation

OPEN ONLY AFTER a saved genuine attempt.

On the clean customer dimension, a many-to-one left merge should preserve accepted order count and amount. The duplicate-key dimension should raise a merge-cardinality failure when validate="many\_to\_one" is used. A successful expanded merge is a defect, not proof.

Changed retry: Create a small product dimension with one duplicate key and prove a many-to-one merge guard fails before aggregation; then repair the dimension and reconcile.

Move on only when the changed retry works with this review closed and you can explain the failure/validation path.

## Review DE05-07 | Data validation: record, batch and relationship rules

OPEN ONLY AFTER a saved genuine attempt.

Record checks parse one row; batch checks include uniqueness/reconciliation; relationship checks span datasets. Pydantic can help with record models, but cannot by itself prove batch uniqueness or foreign-key existence.

Changed retry: Add a syntactically valid customer\_id that does not exist in the dimension. Show record validation passes while relationship validation fails.

Move on only when the changed retry works with this review closed and you can explain the failure/validation path.

## Review DE05-08 | HTTP and REST APIs from zero

OPEN ONLY AFTER a saved genuine attempt.

/health is unauthenticated JSON 200. /v1/\* endpoints require the fictional bearer token. /bad-content is deliberately 200 text/plain. A 200 status alone is not usable-data proof.

Changed retry: Add a local endpoint/fixture expectation where status is 200 but required field is absent. Explain why transport succeeded while contract validation failed.

Move on only when the changed retry works with this review closed and you can explain the failure/validation path.

## Review DE05-09 | Calling APIs safely with requests

OPEN ONLY AFTER a saved genuine attempt.

Safe sequence: timeout -> status policy/raise\_for\_status -> media type -> parse -> payload shape. A 404 is an HTTP client error, a connection refusal is transport failure, and 200 text/plain is a response-contract failure for a JSON endpoint.

Changed retry: Call a path returning 404 and prove your client does not misclassify it as JSON success or transient server failure.

Move on only when the changed retry works with this review closed and you can explain the failure/validation path.

## Review DE05-10 | Pagination and authentication without leaking secrets

OPEN ONLY AFTER a saved genuine attempt.

The simulator has 7 API orders. With page sizes 2 or 4, final fetched count and unique ID count must both equal 7 and match total. Never print/log the Authorization token.

Changed retry: Fetch the same seven records with a different page size and order of page requests under a controlled replay; prove logical completeness/uniqueness remains the same.

Move on only when the changed retry works with this review closed and you can explain the failure/validation path.

## Review DE05-11 | Retries, rate limits and partial failure

OPEN ONLY AFTER a saved genuine attempt.

After /reset, /v1/flaky returns one 500 then 200; /v1/rate-limit returns one 429 with Retry-After=1 then 200. A bad/missing token returns 401 and should not be blindly retried as a transient failure.

Changed retry: Change retry budget from 4 to 2 and explain what evidence would show final failure. Do not create an infinite retry loop.

Move on only when the changed retry works with this review closed and you can explain the failure/validation path.

## Review DE05-12 | Python + PostgreSQL with SQLAlchemy

OPEN ONLY AFTER a saved genuine attempt.

Use Engine.begin() or an explicit transaction boundary for grouped writes; use text() with bound parameters. Learner PostgreSQL evidence is preferred because M03 established PostgreSQL. SQLite QA only proves the Python/SQLAlchemy path, not PostgreSQL-specific runtime.

Changed retry: Insert a different two-row dataset inside one transaction, deliberately violate a unique key, and prove the transaction boundary behavior.

Move on only when the changed retry works with this review closed and you can explain the failure/validation path.

## Review DE05-13 | Configuration, environment variables and secret hygiene

OPEN ONLY AFTER a saved genuine attempt.

Required config should fail near startup. Real secrets do not belong in source, .env.example, saved logs or screenshots. Safe operational values such as batch date/page size may be logged if policy allows.

Changed retry: Switch the input directory via environment variable without editing program logic; then remove one required variable and prove failure occurs before outputs.

Move on only when the changed retry works with this review closed and you can explain the failure/validation path.

## **Review DE05-14 | Logging, testing and reusable pipeline code**

OPEN ONLY AFTER a saved genuine attempt.

Tests should assert invariants. Logs should include safe run context and counts, not secrets. Re-running the same source must not append duplicate logical rows. A second successful run with doubled output is an idempotency failure.

Changed retry: Write a test for invalid date and one for duplicate ID policy. Break one rule intentionally, observe failure, repair only that rule and rerun.

Move on only when the changed retry works with this review closed and you can explain the failure/validation path.

## **Review DE05-15 | Integrated Python data-engineering pipeline**

OPEN ONLY AFTER a saved genuine attempt.

Do not evaluate only the final CSV. Check exact source identity, raw/accepted/rejected reconciliation, accepted-key uniqueness, every rejection reason, accepted amount, failure behavior and rerun behavior. Then close this review before the fresh inventory retest.

Changed retry: After reviewing your weak points, close all review material and solve M05\_Fresh\_Checkpoint\_TODO.py from the fresh inventory fixture. Keep it separate from the first attempt.

Move on only when the changed retry works with this review closed and you can explain the failure/validation path.